Dataset / Knowledge Base Document

1. Title and description

'RAG Study Guide - synthetic sample corpus': 9 short English documents explaining retrieval-augmented generation, chunking, embeddings, FAISS vector search, retrieval metrics, prompting/grounding, prompt-injection security and lexical vs dense retrieval. It is the knowledge base queried by the application (data/knowledge_base/).

2. Origin, licensing and synthetic status

All documents are SYNTHETIC: original text written for this project by the project's AI coding assistant (generated by scripts/kb_content.py and rendered by scripts/build_sample_kb.py). No document was downloaded or copied, so there are no source URLs or access dates. The text summarises well-known published ideas in simplified form (Lewis et al. 2020; Reimers & Gurevych 2019; Johnson et al.; Deerwester et al. 1990; Robertson & Zaragoza 2009; Cormack et al. 2009; OWASP LLM01) and must not be treated as an authoritative source. No third-party licence applies; the team should decide the licence for the released text and should review the content before adopting it as its own. Document 09 is a deliberately poisoned test fixture (it contains an embedded instruction) used only for security tests.

3. Formats, size and processing results (from the executed ingestion)

File	Format	Pages/ segments	Chunks	Chars (clean)	Hdr/ftr lines removed	Status
01_rag_overview.pdf	PDF	2	7	2728	4	ok
02_text_chunking.md	Markdown	5	5	1631	0	ok
03_embeddings_and_similarity.docx	DOCX	5	5	1683	0	ok
04_vector_search_faiss.pdf	PDF	2	6	2066	4	ok
05_retrieval_metrics.md	Markdown	5	5	1469	0	ok
06_prompting_and_grounding.docx	DOCX	5	5	1609	0	ok
07_security_prompt_injection.md	Markdown	4	4	1205	0	ok
08_lexical_vs_dense_retrieval.txt	TXT	5	4	1201	0	ok
09_index_maintenance_note.txt	TXT	4	4	606	0	ok
TOTAL	4 formats	37	45	14198	8	9 ok / 0 failed

Documents processed: 9 of 9; chunks indexed: 45; duplicates removed: 0; chunk length 42-491 characters (mean 322); index build time 2.4 s. PDF 'segments' are pages; for DOCX/MD/TXT they are heading sections (these formats have no reliable page numbers).

4. Preprocessing, cleaning and chunking summary

- Extraction: pypdf per page (PDF), python-docx with Heading/Title styles (DOCX), '#' headings (MD), underlined headings (TXT).
- Cleaning: repeated header/footer and page-number removal (PDF), Unicode NFKC, control characters, de-hyphenation, whitespace collapse, empty-segment removal.
- Chunking: sentence-aware, 500 characters, 100 overlap, per page/section, minimum 40 characters (1 title-only fragment in 08_*.txt was skipped).
- Metadata kept per chunk: chunk id, source file, page (PDF), section heading, character count and text hash.

5. Evaluation data

data/evaluation/eval_set.json holds 34 author-written questions in 8 categories with reference answers and evidence phrases that define relevant chunks (7 are unanswerable or injection-only). They were written with the corpus in view.

6. Known data limitations

- Very small (about 14 thousand characters) and single-style, so retrieval is easier than on real corpora and results do not generalise.
- Synthetic and simplified; not a substitute for the cited literature. English only; no tables, figures, or scanned pages.
- Page numbers exist only for PDFs; the PDF header/footer text is synthetic and chosen to exercise the cleaner.
- Evidence-phrase relevance labels favour exact wording; a chunk could be useful without containing the phrase.